Lateral Automation Techniques and Technologies for Autonomous Vehicles

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***Abstract*—Autonomous driving systems rely on precise steering mechanisms to maintain stability and ensure safe vehicle movement. Among the various control components, estimating the steering angle is essential for functions such as lane alignment, trajectory management, and obstacle response. Conventional rule-based and control-theory approaches have been applied for years, but their effectiveness decreases in unpredictable and highly dynamic road conditions. As a result, data-driven techniques, particularly those based on machine learning, have become a preferred solution. This paper explores different steering angle prediction methods, including regression-based models, convolutional neural networks, and reinforcement learning strategies, as well as vision-oriented, sensor-driven, and integrated hybrid frameworks. The study also discusses how IoT-enabled devices such as cameras, inertial measurement units, GPS modules, and on board processor support continuous data acquisition and real-time decision-making. Existing research is analysed with respect to model structures, datasets, hardware implementations, and evaluation criteria. Additionally, ongoing challenges such as computational efficiency, robustness across diverse driving scenarios, sensor inaccuracies, and safety considerations are addressed, followed by potential future directions emphasizing advanced sensor fusion, edge intelligence, and affordable autonomous steering solutions.**

Keywords- Autonomous vehicles, Steering angle prediction, Lateral vehicle control, IoT-enabled perception.

I. INTRODUCTION

Autonomous driving technology has seen significant advancements over the last decade, mainly propelled by improvements in artificial intelligence, embedded computing, and intelligent sensing systems. A crucial aspect of this technology is lateral automation, which ensures stable steering control and lane alignment, thus enhancing the safety and reliability of vehicle operation. Accurate lateral control allows vehicles to navigate trajectories, avoid road boundaries, and adapt to dynamic environments. As Advanced Driver Assistance Systems (ADAS) move towards greater autonomy, there is an increasing demand for robust steering automation frameworks.

Lateral control utilized rule-based algorithms and classical control theory, relying on explicit lane geometry and vehicle dynamics. While effective in structured environments, these models struggled with real-world variability, such as changes in illumination, occlusions, road wear, and unpredictable traffic. The advent of large-scale datasets, computational enhancements, and sensor-rich platforms has facilitated a transition to data-driven approaches. Techniques such as machine learning, deep neural networks, and reinforcement learning now enable systems to learn steering behaviours from visual and sensor data, enhancing adaptability and decision-making.

Lane perception is fundamental to lateral automation, with recent strides in computer vision yielding various strategies for lane detection, including segmentation-based and parametric approaches. Steering angle prediction models utilize convolutional neural networks and reinforcement learning to transform perceptive outputs into actionable control signals. The integration of diverse sensing modalities, like cameras, LiDAR, radar, and inertial measurement units, improves environmental awareness

through sensor fusion, mitigating individual sensor drawbacks.

Despite advancements, challenges in lateral automation persist, particularly regarding real-time processing, computational efficiency, robustness across varied conditions, and safety validation. Furthermore, research often isolates perception, control, or sensing, indicating a need for a holistic approach that interlinks these domains. This paper proposes a comprehensive review of lateral automation techniques and supporting technologies for autonomous vehicles, covering the entire perception-to-control pipeline—from lane detection and steering prediction frameworks to deep learning, reinforcement learning techniques, sensor technologies, and sensor fusion strategies. The aim is to consolidate current advancements, recognize strengths and weaknesses, and outline future research directions for scalable and dependable steering automation.

II. Lane Detection System

The Advanced Driver Assistance Systems (ADAS) and autonomous driving have advanced quickly due to the increasing use of artificial intelligence in the automotive industry. Real-time vehicle positioning, driving trajectory planning, lane-keeping assistance, and adaptive cruise control are all made possible by lane detection, a fundamental perception task [1]. The majority of lane detection systems currently in use rely on conventional machine learning methods, which are frequently restricted to simple classification tasks rather than intricate feature extraction [2]. Current monocular lane detection methods identify lanes on roads with the help of a camera [3]. Conventional lane detection techniques typically follow a four-step process: image pre-processing, feature extraction, line detection and fitting, and post-processing. These techniques rely on manually created low level features such as colour, gradient, and ridge features. In the previous decade, numerous techniques based on deep neural network have been developed with good performance thanks to advances in deep learning algorithms and computing power [4], [5]. As a type of instance segmentation, lane detection requires that individual pixels be linked to unique lane instances. Conditional instance segmentation techniques have been investigated for lane detection and have demonstrated encouraging results in recent studies. However, because lane markings have distinct structural and geometric features, directly applying these strategies to lane detection is still ineffective [1]. Current approaches can be categorized into four groups based on the line shape description strategy: segmentation-based methods, anchor-based methods, row-wise detection methods, and parametric prediction methods [1], [3].

A. Segmentation Based Methods

Among the most popular techniques for lane detection, segmentation based approaches have proven to be effective in real world scenarios. Many of these techniques use a post clustering to increase instance level accuracy.[1], [6]. These methods group pixels that correspond to lane markings into distinct lane instances after first classifying each pixel in the image using binary segmentation. In order to decode the final lane representations, the resulting instance-level segmentation maps are usually refined and then subjected to post-processing procedures [3], [7]. Many of the earlier methods treat lane detection simply as a pixel level classification task, overlooking the fact that lanes are spatially continuous structures with geometric properties that should be considered during detection [3], [6]. To overcome this problem, anchor-based methods and row-wise detection methods are developed. [1], [3]. Jia-Qi Zhang et al. [3] propose HoughLaneNet, a segmentation-based lane detection method that incorporates Discrete Hough Transform (DHT) based feature aggregation to improve robustness under weak visual cues such as occlusion and lane wear. Unlike traditional segmentation approaches that require multi class instance separation, their method only predicts a binary lane map and performs lane instance discrimination directly in the Hough parameter space. Multi-scale features extracted using a ResNet backbone and FPN are transformed into Hough space, where lane features are aggregated and selected. These features are then fused with spatial features to predict individual lanes as sequences of 2D points, resulting in a simpler yet effective one-stage framework.

B. Anchor based Methods

Xu et al. [8] introduced an anchor-based lane detection approach that takes advantage of the inherent structural properties of lanes. Using vertical anchors, the method predicts point-wise offsets to reconstruct each lane, and its effectiveness was demonstrated on the CurveLanes dataset. Anchor based lane detection methods follow a top to down pipeline and focus on modelling lane geometry by directly regressing relative coordinates along the lane shape. These approaches reduce ambiguity in regions with weak or missing visual cues and improve instance level lane discrimination. Since lane markings are thin and elongated, conventional box shaped anchors used in object detection are not suitable. Instead, vertical line anchors or slender, lane shaped anchors are commonly adopted to better capture lane structure [1], [9]. Anchor based approaches incorporate prior geometric knowledge of lane structure directly into the model, which helps achieve robust predictions even under heavy occlusion. However, their detection accuracy remains sensitive to how finely the anchors are defined [3].

C. Row wise Detection Methods

Explainability Row wise lane detection methods effectively exploit prior knowledge about lane shape by predicting the lane position independently for each image row. During training, global shape consistency is enforced through constraints applied to the predicted location at each row. By encouraging continuity and coherence across successive rows, these approaches implicitly model lane geometry and produce smooth, structurally consistent lane predictions [1], [10], [11]. Motivated by the strong balance between accuracy and computational efficiency achieved by row wise lane detection methods, Lizhe Liu [1] adopts a row wise formulation and introduces several novel strategies to address challenges related to instance-level lane

discrimination. Lizhe Liu et al.[1] propose CondLaneNet, a top-down approach that detects lane instances before refining their shapes. By using conditional convolution and a row-wise formulation, the method effectively addresses instance-level ambiguity, while an interaction module helps manage complex lane layouts. The approach achieves strong performance across major benchmarks and maintains real-time efficiency in its lightweight form.

D. Parametric Prediction Methods

Unlike the previously discussed approaches that predict discrete lane points, parametric prediction methods directly model lanes as continuous curves defined by parametric equations [1]. Tabelini et al.[12] introduced a parametric lane representation in which lanes are modelled as polynomial curves, with the network directly regressing the corresponding curve parameters. Using third-order polynomials, their approach achieved competitive performance, reaching approximately 90% accuracy on the TuSimple dataset. Despite these promising results, parametric prediction methods have generally not outperformed other lane detection approaches in terms of overall accuracy [1], [3]. With respect to the polynomial order used for lane representation, the marginal accuracy difference observed when employing lower-order polynomials highlights the inherent imbalance of the dataset. Notably, reducing the model to first-order polynomials resulted in an accuracy drop of only 0.35 percentage points [12].

TABLE I. METHODS FOR LANE DETECTION.

Method / Model	Core Idea	Contribution
CondLaneNet [1]	Conditional convolution with row-wise formulation	Detects lane instances first, then refines shapes
YOLO-based Lane Detection [2]	Object detection approach	Real-time lane localization
HoughLaneNet [3]	Segmentation + Hough Transform	Hough-space instance separation
Hybrid Spatial–Temporal [4]	Spatial and temporal learning	Improves temporal consistency
LVLane [5]	Robust deep learning	Handles adverse conditions
Self-Attention Distillation [6]	Lightweight CNN learning	Efficient compact models
SCNN [7]	Spatial message passing	Captures lane continuity
CurveLane-NAS [8]	Anchor-based + NAS	Lane-sensitive architecture
LaneATT [9]	Attention-guided anchors	Improves anchor interaction
Ultra Fast Lane Detection [10]	Row-wise prediction	Extremely fast
FastDraw [11]	Sequential prediction	Handles rare lane cases
PolyLaneNet [12]	Polynomial regression	Parametric lane modeling

III. STEERING ANGLE PREDICTION

Lane keeping is a core lateral control function that automatically steers an autonomous vehicle to remain within lane boundaries, even under varying road conditions and external disturbances. This is achieved by estimating the desired steering angle from the vehicle's lateral dynamics, which serves as the primary input to the lateral controller for executing stable and accurate lane keeping maneuvers [13]. In autonomous vehicle steering control, vehicle motion is primarily governed by two components: lateral and longitudinal dynamics. Steering specifically addresses lateral motion, with the objective of regulating the vehicle's position within the lane and ensuring stable and accurate path following [14]. Steering angle prediction methods for autonomous vehicles can generally be divided into two main categories. The first category relies on explicitly extracting road or lane boundary information and then applying mathematical or statistical models to compute the steering angle. In contrast, the second category is based on imitation learning, where artificial neural networks learn steering behaviour directly from expert demonstrations. In this approach, lane or boundary features are not manually extracted; instead, the model learns the mapping from visual inputs such as single images or sequences of images to steering angle outputs. In both approaches, the predicted steering angle is combined with other lateral control parameters to ensure that the autonomous vehicle remains safely within road boundaries [13].

A. Steering Angle Prediction Using Computer Vision

In this steering angle prediction approach, the process begins with explicitly extracting lane markings or road boundary coordinates. Image frames are captured in real time using a camera mounted on the vehicle and are then pre-processed to enhance important road features and reduce noise. Based on these refined images, lane boundaries are detected and used to compute the required steering angle. To improve robustness in challenging situations such as occlusions, faded lane markings, or illumination changes, some studies further apply lane tracking across consecutive frames. This temporal consistency is achieved using filtering techniques such as Kalman filters, particle filters, or Bayesian filters, allowing reliable road detection even when lane information is partially missing or noisy [13]. Traditionally, the pre-processing stage relies on a set of image enhancement techniques to simplify the visual input. These commonly include reducing image resolution through down sampling, segmenting the image to isolate relevant features, smoothing to suppress noise, selecting a Region of

Interest that focuses on the road area, and applying Inverse Perspective Mapping (IPM) to obtain a clearer representation of lane geometry [13], [14]. The steering angle is estimated by finding the intersection of the detected left and right lane boundaries. If this point aligns with the road centre then no steering adjustment is needed. Otherwise, a central reference path is formed using the lane geometry and the steering angle is computed from the deviation between this path and the vehicle's current heading [13].

B. Steering Angle Prediction using Deep Learning

Deep learning has significantly improved steering angle prediction in autonomous vehicles by enabling models to learn directly from raw, unlabelled data. By analysing the overall scene context and focusing on relevant features at multiple levels of detail, deep learning methods can effectively capture both fine-grained and high-level visual information needed for accurate steering decisions [14]. This section reviews the deep learning architectures used for steering angle prediction in autonomous vehicles. Aires Da Conceicao [15] proposed a steering and road direction prediction system based on the NVIDIA end to end convolutional neural network architecture. The system uses road images captured in real time by a camera mounted on the vehicle as input to learn the relationship between visual features and driving direction. The method follows a supervised learning approach in which labelled image data are used to train a convolutional neural network model. After training, the model predicts the road direction directly from camera images and produces a continuous output that supports steering control in autonomous vehicles. Rodolfo Valiente [16] addressed steering angle prediction as a regression problem by directly mapping visual input to continuous steering values. The study highlights that using a sequence of images rather than a single frame overcome challenges such as strong sunlight, image noise, or dead end roads. Temporal relationships between consecutive frames help the system infer more reliable steering decisions. To model these temporal dependencies, Valiente employed Long Short Term Memory networks, which retain relevant information from previous frames and use it to predict the next steering angle. The proposed approach further extends this idea by incorporating vehicle to vehicle communication. Images captured by a leading vehicle are shared with a following vehicle and combined with locally captured images, allowing the system to anticipate road conditions beyond the current field of view. The model uses an end to end deep learning framework combining CNN, LSTM, and fully connected layers to predict steering angles. CNN extracts visual features, LSTM captures temporal information across frames, and the final layers output a continuous steering value using regression with mean squared error loss [13], [14], [16].

C. Steering Angle Prediction using Deep Reinforcement Learning

Deep Reinforcement Learning combines deep learning and reinforcement learning. It has enabled solutions to complex tasks that were difficult for traditional machine learning methods. In this approach an agent learns through trial and error by interacting with its environment and receiving rewards with the goal of maximizing long term returns over time [14]. Qadeer Khan [17] proposed a deep reinforcement learning based steering control framework that combines perception and decision making in an end to end manner. In this approach, images from a front facing camera are first processed by a perception model to generate a compact semantic representation of the driving scene. Rather than using raw images directly, the reinforcement learning agent is trained on this latent semantic embedding, which reduces complexity and improves learning efficiency. The steering controller is trained through interaction with the environment using a reward driven trial and error process instead of supervised labels. Positive rewards are assigned for safe lane following, while negative rewards are given for unsafe behaviours such as lane departure, off road driving, or collisions. Training is carried out in the CARLA simulator, where the agent learns appropriate steering actions by maximizing long term cumulative rewards. To address the large continuous state space, the study adopts a model free Deep Q Learning strategy, where a neural network approximates state action values instead of relying on explicit environment models. Techniques such as experience replay and a separate target network are used to stabilize training. By operating on semantic latent states rather than full images, the method significantly reduces training time while maintaining effective steering control performance. Keyu Wu [18] introduced a deep reinforcement learning based steering control framework called BND-DDQN, designed to model complex nonlinear driving policies. The approach is built on a Double Deep Q Network architecture enhanced with branching, dueling, and noisy components to improve learning stability and exploration efficiency. The model processes a short sequence of depth images using convolutional layers to extract compact feature representations, which are then shared across multiple decision branches. These branches separately estimate the overall state value and the action advantages associated with linear and angular vehicle motions. By separating these components, the network is able to learn more precise control policies for steering and speed. To encourage effective exploration during training, the framework replaces traditional random exploration strategies with noisy neural networks, where controlled noise is injected directly into network parameters. This allows the agent to explore the environment in a more structured and consistent manner. The estimated state values and action advantages are combined to produce final action value predictions, and actions are selected greedily based on the learned policy without relying on epsilon greedy exploration.

TABLE II. STEERING ANGLE PREDICTION TECHNIQUES.

Method	Model / Algorithm	Dataset / Platform	Key Contribution
Deep Learning	CNN (NVIDIA end to end)	Real world images	End to end steering angle prediction

Method	Model / Algorithm	Dataset / Platform	Key Contribution
(Imitation Learning) [15]			from road images
Deep Learning (CNN + LSTM) [16]	CNN + LSTM + FC	Simulation and V2V	Temporal steering prediction using sequential images and cooperative perception
Deep Reinforcement Learning [17]	DQN based RL	CARLA simulator	Latent space reinforcement learning for steering control
Deep Reinforcement Learning [18]	BND DDQN	Simulation	Branching dueling noisy DDQN for stable nonlinear steering control

IV. Sensor Technology for Lateral Automation

Sensors are essential components of autonomous vehicles as they convert physical changes in the vehicle or its surroundings into numerical data that can be processed by perception and control systems. Based on their operational principle sensors are generally classified into internal state and external state sensors. Internal state sensors also known as proprioceptive sensors monitor the dynamic condition of the vehicle by measuring variables such as angular rate wheel speed force and battery status. Typical examples include inertial measurement units encoders gyroscopes magnetometers and global navigation satellite system receivers. In contrast external state sensors referred to as exteroceptive sensors capture information from the surrounding environment including object distance road features and light intensity. Common exteroceptive sensors used in autonomous vehicles include cameras radar LiDAR and ultrasonic sensors. Sensors may further be categorized as active or passive depending on whether they emit energy into the environment. In autonomous driving applications the integration of these sensors is essential for environment perception obstacle detection and safe path planning [19], [20].

A. Cameras

Camera based vision sensors operate on optical imaging principles in which incoming light is focused through lenses onto image sensors such as CCD or CMOS. CCD sensors transfer charge sequentially which results in high image quality but higher power consumption. In contrast CMOS sensors integrate signal amplification and readout within each pixel enabling higher resolution images with rich colour texture and shape information. As a result camera sensors are widely used for semantic perception tasks including lane detection traffic sign recognition and pedestrian classification. Cameras are compact and cost effective which makes them attractive for large scale deployment. However they lack direct depth perception and their performance is sensitive to poor ligh zting conditions shadows and occlusions [21]. Stereo cameras, often called binocular cameras, replicate biological depth perception by using the difference between two slightly offset images to infer distance. Each system consists of two image sensors separated by a fixed baseline, which is the physical distance between the sensors and varies by camera design. Orbbec 3D cameras commonly used in autonomous and intelligent vehicle applications employ a baseline of about 75 mm. By analysing image disparity between the paired views, stereo cameras can generate depth information through geometric triangulation. This depth output is widely used in robotics, where stereo vision pipelines, such as those available in the Robot Operating System, support tasks including simultaneous localization and mapping and autonomous navigation [20].

B. LiDAR

A key perception sensor for contemporary Advanced Driver Assistance Systems and autonomous cars, LiDAR (Light Detection and Ranging) is a remote sensing technology that began with aerial and terrain mapping applications. The technology allows for accurate distance estimation by generating laser pulses and timing the return of reflected signals from nearby objects. LiDAR continuously scans the environment to create a point cloud, a 3D representation of the scene that helps autonomous driving systems perceive the environment accurately and comprehend space [20]. Camera sensors are affordable but they do not provide accurate depth information which limits their effectiveness for precise three dimensional object detection. Their performance also degrades in challenging weather conditions such as fog rain and snow. Sensors that produce point cloud data such as LiDAR and radar overcome these limitations by directly measuring depth which enables more reliable estimation of object size and position. However point cloud data are sparse irregular and unevenly distributed which makes them unsuitable for direct use with conventional image based deep learning methods. As a result specialized processing techniques are required to handle this type of data and to reduce sparsity related issues. In autonomous driving research point cloud representations are generally grouped into projection based voxel based and raw point cloud approaches each offering different trade-offs between accuracy and computational efficiency [22].

C. Radar

Radar is a long established sensing technology that detects surrounding objects by transmitting electromagnetic waves and analysing their reflections. The time taken for the

signals to return is used to estimate the distance to targets while changes in the signal frequency caused by relative motion enable the estimation of object speed. This combined ability to measure range and velocity makes radar well suited for identifying and tracking moving obstacles in automotive environments [20], [21]. Radar datasets used in autonomous driving research differ significantly in terms of sensor type and data representation. Radar sensors are commonly categorized into low resolution and high resolution systems, with high resolution achieved through various technical approaches such as advanced antenna configurations, cooperative sensing, and synthetic or extended aperture techniques. Commercial automotive radars typically output point cloud data containing range azimuth Doppler velocity and radar cross section information, while newer four dimensional radars additionally provide elevation measurements. Some experimental radar systems also offer access to raw radar data including digitized signals and intermediate representations. In autonomous driving applications radar is mainly employed for localization and object detection. While detection remains the primary focus in many studies localization datasets are also relevant since they often include synchronized radar LiDAR and camera data which can be repurposed for detection tasks. Radar annotations are provided at different levels of detail depending on the dataset. These range from two dimensional bounding boxes in bird's eye view with orientation information to full three dimensional bounding boxes that additionally encode object height and orientation [23].

D. Inertial Measurement Unit

Inertial navigation also called inertial odometry estimates the position orientation and three dimensional velocity of a moving system using data from an Inertial Measurement Unit. The IMU contains accelerometers and gyroscopes and enables real time motion estimation through dead reckoning without relying on external references. While inertial navigation has been studied for many years the availability of small and low cost sensors has increased its use in robotics and autonomous systems. High precision inertial navigation systems can provide very accurate localization but they are expensive and require complex initialization. In contrast low cost IMUs are more practical for autonomous platforms but suffer from significant errors such as sensor noise bias and drift. These errors accumulate over time and lead to rapid degradation in localization accuracy. As a result inertial navigation alone is not sufficient for long term autonomy and is commonly combined with other sensors such as GNSS cameras or LiDAR to improve reliability and accuracy [24] [19]. Martin Brossard [24] presents a long-term localization approach for wheeled robots that relies solely on inertial measurements. The method takes advantage of characteristic motion patterns commonly observed in wheeled vehicles, which provide valuable information about the system's motion when correctly identified. These motion patterns are detected automatically in real time using only IMU data through a learning-based model trained on inertial measurements and ground truth trajectories. The detected motion information is integrated into an advanced Kalman filtering framework, where it is treated as additional constraints and statistically combined with raw IMU measurements. This enables accurate estimation of vehicle pose, velocity, and sensor biases, along with their associated uncertainty. Experimental results on a public vehicle dataset show that the proposed IMU-only approach achieves high long-term accuracy over extended driving sequences. The method outperforms conventional wheel odometry as well as wheel odometry aided by high-grade gyroscopes with significantly lower drift. These findings demonstrate that identifying meaningful patterns in inertial data and incorporating them into a principled filtering framework can significantly enhance localization. Moreover, the approach is flexible and can be extended to include measurements from other sensors, making it a robust building block for broader autonomous localization systems.

E. Sensor Fusion

Sensor fusion is a fundamental component of autonomous systems such as self-driving cars and unmanned ground vehicles. It combines data from multiple sensors to reduce uncertainties and address the limitations of individual devices while providing a more accurate and consistent understanding of the environment. By integrating complementary modalities such as camera and radar autonomous systems can obtain both high resolution visual information and measurements of obstacle velocities which improves situational awareness and supports reliable navigation in diverse conditions [20].

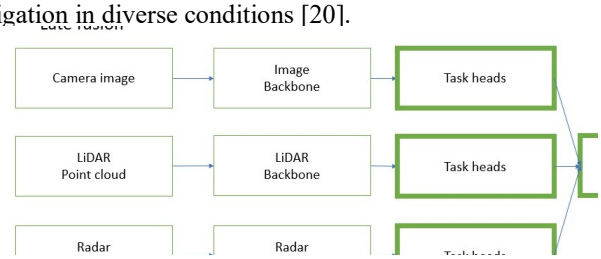

Figure 1. Late Sensor Fusion Flow

Fusion in autonomous driving systems are divided into three stages: early fusion, mid-level fusion, and late fusion. Each stage reflects the timing and abstraction level at which sensor data from cameras, LiDAR, Radar, and sometimes IMU or GPS, is integrated within the processing pipeline, Early fusion combines raw data from multiple sensors before feature extraction allowing the system to learn joint representations from the start but it requires large models and can be affected by differences in sensor resolution coordinate frames sampling rates and noise. Mid-level fusion extracts meaningful features from each sensor independently before combining them which reduces input size mitigates alignment issues and can handle missing or degraded data. However it may lose some fine details across sensors. Late fusion merges outputs at the decision level providing robustness modularity and computational efficiency but it is less effective for tasks that require detailed spatial or semantic integration [25]. The figure 1 shows the flow in the late fusion.

V. CONCLUSION

Autonomous steering control has become a fundamental component in the progress of self-driving vehicle technology, particularly with the integration of machine learning and IoT-based systems. The review of existing research shows a clear transition from conventional rule-based and control-theoretic methods to data-driven approaches capable of learning steering behaviour directly from visual and sensor inputs. Techniques such as convolutional neural networks have demonstrated strong capability in predicting steering angles with reasonable accuracy, while embedded platforms like Arduino and related IoT modules enable practical real-time actuation at a lower cost. Although high-end systems equipped with advanced sensors achieve high precision, their complexity and expense limit accessibility for academic and small-scale experimentation. Low-cost implementations illustrate that effective steering automation can still be achieved using optimized models and modular hardware integration. However, challenges related to real-time inference, robustness under varying environmental conditions, sensor noise, and operational safety continue to require careful attention. Overall, the combination of efficient learning algorithms and scalable IoT hardware presents a promising pathway toward affordable, adaptable, and research-oriented autonomous steering solutions.

ACKNOWLEDGMENT

The author gratefully acknowledges A. P. Shah Institute of Technology, Thane for providing research environment and support in publication of research.

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